

CHAPTER II.3, WEEK 4

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3.10. *Fibres of a morphism.*

- (a) If $f : X \rightarrow Y$ is a morphism and $y \in Y$ a point, show that $\text{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.
- (b) Let $X = \text{Spec } k[s, t]/(s - t^2)$, $Y = \text{Spec } k[s]$, and $f : X \rightarrow Y$ the morphism defined by $s \mapsto s$. If $y \in Y$ is a point $a \in K$ with $a \neq 0$, show that the fibre X_y consists of two points, with residue field k . If $y \in Y$ corresponds to $0 \in k$, show that the fibre X_y is a nonreduced one-point scheme. If η is the generic point of Y , show that X_η is a one-point scheme, whose residue field is an extension of degree two of the residue field of η . (Assume k algebraically closed.)

Proof. (a) We begin by considering the affine case, $Y = \text{Spec } A$, $X = \text{Spec } B$, and $f^\#(Y) = \varphi : A \rightarrow B$. An identical method to that of Ex. 2.1 gives us the following lemma.

Lemma 1. *Let A be a ring and $\phi : A \rightarrow S^{-1}A$ the inclusion into its localization. Then $\phi^* : \text{Spec}(S^{-1}A) \rightarrow \text{Spec } A$ is a homeomorphism onto its image in $\text{Spec } A = X$. We denote this image $S^{-1}X$.*

The next lemma is easy, but worth proving.

Lemma 2. *Let $f : A \rightarrow B$ be a ring morphism, $X := \text{Spec } A$, $Y := \text{Spec } B$, and S a multiplicative subset of A . Then denoting $S^{-1}B = f(S)^{-1}B \cong B_A \otimes_A S^{-1}A$ and identifying spectra as in the prior lemma,*

$$S^{-1}f^* : \text{Spec}(S^{-1}B) \rightarrow \text{Spec}(S^{-1}A)$$

is the restriction of f^ to $S^{-1}Y$, and $S^{-1}Y = f^{*-1}(S^{-1}X)$.*

Proof. We have the following commutative diagram.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \phi_A \downarrow & & \downarrow \phi_B \\ S^{-1}A & \xrightarrow{S^{-1}f} & S^{-1}B \end{array}$$

Applying the contravariant functor Spec , we get a diagram of topological spaces.

$$\begin{array}{ccc} X & \xleftarrow{f^*} & Y \\ \phi_A^* \uparrow & & \uparrow \phi_B^* \\ \text{Spec}(S^{-1}A) & \xleftarrow{S^{-1}f^*} & \text{Spec}(S^{-1}B) \end{array}$$

Then

$$f^*(S^{-1}Y) = \phi_A^*(S^{-1}f^*(\text{Spec}(S^{-1}B))) \subset S^{-1}X$$

allows us to restrict to the following diagram.

$$\begin{array}{ccc} S^{-1}X & \xleftarrow{f^*|_{S^{-1}Y}} & S^{-1}Y \\ \phi_A^* \uparrow & & \uparrow \phi_B^* \\ \text{Spec}(S^{-1}A) & \xleftarrow{S^{-1}f^*} & \text{Spec}(S^{-1}B) \end{array}$$

By the preceding lemma, ϕ_A^* and ϕ_B^* are homeomorphisms onto their image, whence the result follows. $\square$

One last lemma precedes the proof of the desired statement.

Lemma 3. *Let $\mathfrak{a}$ be an ideal of A and let $\mathfrak{b} = \mathfrak{a}B$ be its extension in B . Let $f : A/\mathfrak{a} \rightarrow B/\mathfrak{b}$ be the homomorphism induced by f . If $\text{Spec}(A/\mathfrak{a})$ is identified with its canonical image $V(\mathfrak{a}) \subset X$, and $\text{Spec}(B/\mathfrak{b})$ with $V(\mathfrak{b}) \subset Y$, show that $\bar{f}^* = f^*|_{V(\mathfrak{b})}$.*

Proof. We have the following commutative diagram.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \pi_A \downarrow & & \downarrow \pi_B \\ A/\mathfrak{a} & \xrightarrow{\bar{f}} & B/\mathfrak{b} \end{array}$$

Once again applying Spec ,

$$\begin{array}{ccc} X & \xleftarrow{f^*} & Y \\ \pi_A^* \uparrow & & \uparrow \pi_B^* \\ \text{Spec}(A/\mathfrak{a}) & \xleftarrow{\bar{f}^*} & \text{Spec}(B/\mathfrak{b}) \end{array}$$

and restricting as before gives us our last diagram.

$$\begin{array}{ccc} V(\mathfrak{a}) & \xleftarrow{f^*|_{V(\mathfrak{b})}} & V(\mathfrak{b}) \\ \pi_A^* \uparrow & & \uparrow \pi_B^* \\ \text{Spec}(A/\mathfrak{a}) & \xleftarrow{\bar{f}^*} & \text{Spec}(B/\mathfrak{b}) \end{array}$$

By Exercise 2.18, π_A^* and π_B^* are homeomorphisms onto their images $V(\mathfrak{a})$ and $V(\mathfrak{b})$, so that the result follows. $\square$

Finally, we set $X = \text{Spec } B$, $Y = \text{Spec } A$, and fix $y = \mathfrak{p} \in \text{Spec } A$. Clearly, we have homeomorphisms

$$\begin{aligned} X_{\mathfrak{p}} &= \text{Spec}(B \otimes_A k(\mathfrak{p})) \\ &\cong \text{Spec}(B \otimes_A A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}) \\ &\cong \text{Spec}(B_{A_{\mathfrak{p}}}/\mathfrak{p}B_{A_{\mathfrak{p}}}) \\ &\cong V(\mathfrak{p}B_{A_{\mathfrak{p}}}), \end{aligned}$$

where the last equality follows from Exercise 2.18 as before. Then equalities

$$\begin{aligned}
&\cong V(\varphi_{\mathfrak{p}}(\mathfrak{m}_{\mathfrak{p}})B_{A_{\mathfrak{p}}}) \\
&\cong (\varphi_{\mathfrak{p}}^*)^{-1}(V(\mathfrak{m}_{\mathfrak{p}})) \\
&\cong f_{\mathfrak{p}}^{-1}(\mathfrak{m}_{\mathfrak{p}}) \\
&\cong f^{-1}(\mathfrak{p}).
\end{aligned}$$

follow from minimality of the extension of an ideal, maximality of $\mathfrak{m}_{\mathfrak{p}} \trianglelefteq A_{\mathfrak{p}}$, and the second lemma respectively.¹

Fix affine open covers $\{U_i = \operatorname{Spec} A_i\}$ of X and $\{V_j = \operatorname{Spec} B_j\}$ of Y . Let $g : \operatorname{Spec} k(y) \rightarrow Y$ be the natural map, giving $\operatorname{Spec} k(y)$ the structure of a Y -scheme. We have

$$k(y) = \mathcal{O}_{Y,y}/\mathfrak{m}_y \mathcal{O}_{Y,y} \cong (B_j)_y / y(B_j)_y := k_j(y)$$

for all $y \in Y_j$.

Then X_y is the gluing of the spaces $f^{-1}(Y_j) \times_{Y_j} \operatorname{Spec} k_j(y)$, over all $Y_j \ni y$. We can write

$$X_i \cap f^{-1}(Y_j) = \bigcup_{\substack{g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk})$$

as a union of affines. Defining $f_{ijk} := f|_{D_i(g_{ijk})} : D_i(g_{ijk}) \rightarrow Y_j$, the affine case gives us that

$$\begin{aligned}
X_y &= \bigcup_{\substack{i,j,g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk}) \times_{Y_j} \operatorname{Spec} k_j(y) \\
&= \bigcup_{i,j,k} D_i(g_{ijk})_y \\
&= \bigcup_{i,j,k} f_{ijk}^{-1}(y) \\
&= f^{-1}(y) \cap \bigcup_{i,j,k} D_i(g_{ijk}) \\
&= f^{-1}(y).
\end{aligned}$$

(b) We have that $X_a = \operatorname{Spec} A_a$, where

$$\begin{aligned}
A_a &= \frac{k[s,t]}{(s-t^2)} \otimes_{k[s]} \frac{k[s]_{(s-a)}}{(s-a)k[s]_{(s-a)}} \\
&\cong k[t] \otimes_{k[t^2]} \frac{k[t^2]}{(a-t^2)} \\
&\cong \frac{k[t]}{(a-t^2)}.
\end{aligned}$$

¹I originally wrote this out a while ago, so I hadn't written it up again, let alone in simplified form, when we had our meeting. For some reason I thought (b) was the interesting part lol.

Then $\text{Spec } A_a \cong V(a - t^2) = \{(\sqrt{a} + t, \sqrt{a} - t)\} \subset \text{Spec } k[t]$. If $a \neq 0$, then this is a two point set, with residue fields given by

$$\begin{aligned} k_a(\sqrt{a} - t) &= \frac{k[t]}{(\sqrt{a} - t)} \cong k, \\ k_a(\sqrt{a} + t) &= \frac{k[t]}{(\sqrt{a} + t)} \cong k. \end{aligned}$$

If $a = 0$, then $\text{Spec } A_0 = \{t\}$ is singleton, and $t \in A_0$ is a nilpotent.

Finally, we consider

$$\begin{aligned} A_\eta &\cong \frac{k[s, t]}{(s - t^2)} \otimes_{k[s]} \frac{k[s]_{(0)}}{(0)k[s]_{(0)}} \\ &\cong k[t] \otimes_{k[t^2]} k[t^2] \\ &\cong k(s)[\sqrt{s}]. \end{aligned}$$

Then A_η is a PID, so its only ideals are of the form $(f(\sqrt{s}))$ for $f(\sqrt{s}) = f_1(s) + f_2(s)\sqrt{s}$. Then $f\bar{f}(\sqrt{s}) \in k(s)$ defined in the obvious manner is invertible whenever $f \neq 0$, so $(f(\sqrt{s})) = 0$, A_η gives us that $\text{Spec } A_\eta$ is singleton. Furthermore, $(A_\eta)_{(0)} \cong k(\sqrt{s})$, which is a degree two extension over $k(\eta) = k(s)$, as desired. $\square$

3.12. Closed subschemes of $\text{Proj } S$.

(a) Let $\varphi : S \rightarrow T$ be a surjective homomorphism of graded rings, preserving degrees. Show that the open set U is equal to $\text{Proj } T$, and the morphism $f : \text{Proj } T \rightarrow \text{Proj } S$ is a closed immersion.

(b) If $I \subset S$ is a homogeneous ideal, take $T = S/I$ and let Y be the closed subscheme of $X = \text{Proj } S$ defined as image of the closed immersion $\text{Proj } S/I \rightarrow X$. Show that different homogeneous ideals can give rise to the same closed subscheme. For example, let d_0 be an integer and let $I' = \bigoplus_{d \geq 0} I_d$. Show that I and I' determine the same subscheme.

We will see later that every closed subscheme of X comes from a homogeneous ideal I of S (at least in the case where S is a polynomial ring over S_0).

Proof. (a) Denote $Y = \text{Proj } S$ and $X = \text{Proj } T$. We have $\varphi(S_+) = T_+$, so that $U = X$ by definition. Fix homogeneous $g \in S_+$. Surjectivity of $\varphi : S \rightarrow T$ implies $\varphi_g : S_g \rightarrow T_{\varphi(g)}$ is surjective. Then φ degree preserving gives us that $(S_g)_d = \varphi_g^{-1}((T_g)_d)$ for all $d \geq 0$, whence $\varphi_g : S_g \rightarrow T_{\varphi(g)}$ is surjective. It follows that $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective.

Surjectivity of φ gives us $\varphi(\varphi^{-1}(\mathfrak{b}))$ for all $\mathfrak{b} \trianglelefteq T$, whence

$$\mathfrak{p} \supset \mathfrak{b} \iff \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{b}).$$

Then

$$f(V(\mathfrak{b})) = V(\varphi^{-1}(\mathfrak{b})) \cap f(X)$$

shows that, in particular,

$$f(X) = V(\ker \varphi) \cap f(X) \subset V(\ker \varphi).$$

Take $\mathfrak{p} \in V(\ker \varphi)$, and $a \in \varphi^{-1}(\varphi(\mathfrak{p}))$. Then we have $a' \in \mathfrak{p}$ with $\varphi(a) = \varphi(a')$, whence $a - a' \in \ker \varphi \subset \mathfrak{p}$ implies $a \in \mathfrak{p}$. Thus, $\mathfrak{p} = \varphi^{-1}(\varphi(\mathfrak{p})) \in f(X)$ gives us $f(X) = V(\ker \varphi)$.

We conclude that

$$f(V(\mathfrak{b})) = V(\ker \varphi + \varphi^{-1}(\mathfrak{b}))$$

is closed in $\text{Proj } S$, so that f is a homeomorphism onto the closed set $f(X) = V(\ker \varphi)$.

(b) Let $\pi : S \rightarrow S/I$ and $\pi' : S \rightarrow S/I'$ be the natural projections. Since $I' \subset I = \ker \pi$, we can factor $\pi = \varphi\pi'$, where $\varphi : S/I' \rightarrow S/I$ is the identity $\varphi_d : S_d/I'_d = S_d/I_d \rightarrow S_d/I_d$ for $d \geq d_0$. Then Ex. 2.14(b) gives us that $\varphi^* : \text{Proj}(S/I) \rightarrow \text{Proj}(S/I')$ is an isomorphism of sheaves, so $(\pi')^*(\text{Proj}(S/I')) = \varphi^*\pi^*(\text{Proj}(S/I)) \cong \pi^*(\text{Proj}(S/I))$ give the same closed subscheme structure. $\square$